VU QUIZ REPORT

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Subject: MTH101

Quiz Completed: 1/8/2026, 12:16:40 AM
Total Questions: 9
Mathematical Questions: 1
CROPPED Screenshots: 1

Question 1:

What is the relationship between the area under a curve and the definite integral?

A. The area is the antiderivative of the integral

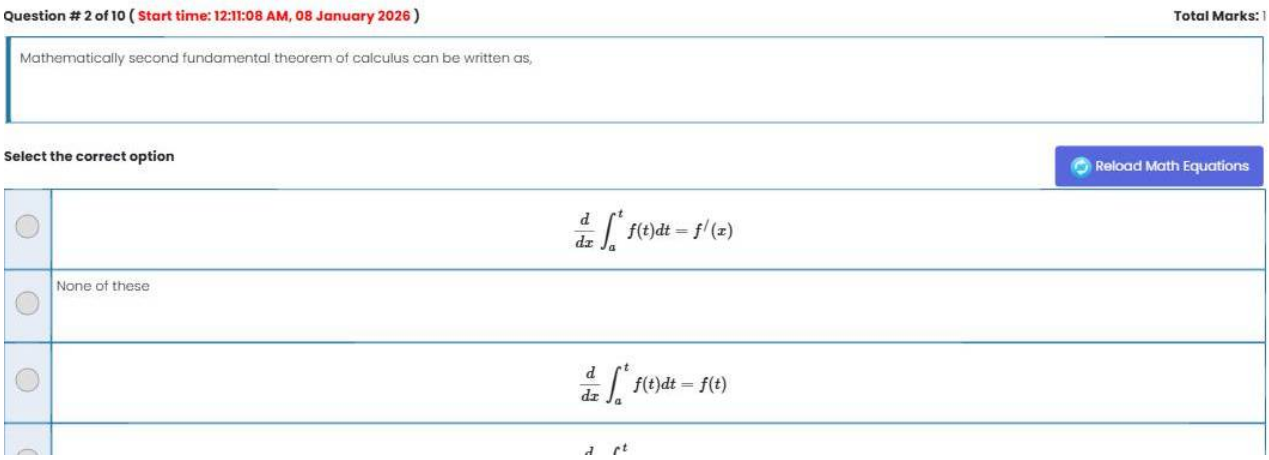
B. The area is the derivative of the integral.

C. Both are same concepts.

D. They are unrelated concepts.

Selected Answer: C

Question 2:



You selected: D

Mathematical question (CROPPED screenshot - top 20% removed)

Selected Answer: D

Question 3:

Suppose two differentiable functions f(x) and g(x) have the same graph on the interval [a, b]. How their arc lengths are related?

A. The arc length of g(x) is greater.

B. The arc lengths are equal.

C. The arc length of g(x) is zero.

D. The arc length of f(x) is greater.

Selected Answer: B

Question 4:

If a similar solid is rotated around the x-axis over a closed interval [a, b] then the corresponding volume of revolution is ....

A. .

B. .

C. .

D. .

Selected Answer: B

Question 10:

.

A. .

B. .

C. .

D. .

Selected Answer: C

Question 6:

If the function f(x) is not smooth or differentiable, how would it affect the surface area calculation?

A. It has no effect.

B. The surface area becomes infinite.

C. The calculation becomes more accurate.

D. The surface area calculation is not defined.

Selected Answer: D

Question 7:

In the disk method, what is the shape of a cross-sections that is perpendicular to the axis of rotation?

A. Rectangular

B. Triangular

C. Circular

D. Square

Selected Answer: C

Question 8:

Arc length of the curve y=1 from x=a to x=b is .....

A. a-b

B. b-a

C. 0

D. 2(b-a)

Selected Answer: B

Question 9:

What is the formula for finding the volume of a cylindrical shell? where R is the radius of the outer cylinder, r is the radius of the inner cylinder and h is the length of the cylindrical shell.

A. .

B. .

C. .

D. .

Selected Answer: A

Mathematical Questions Note:

Questions containing mathematical equations and symbols are displayed as CROPPED screenshots (top 20% removed for privacy) to preserve the original formatting. Your selected answers are displayed below each screenshot for easy reference.